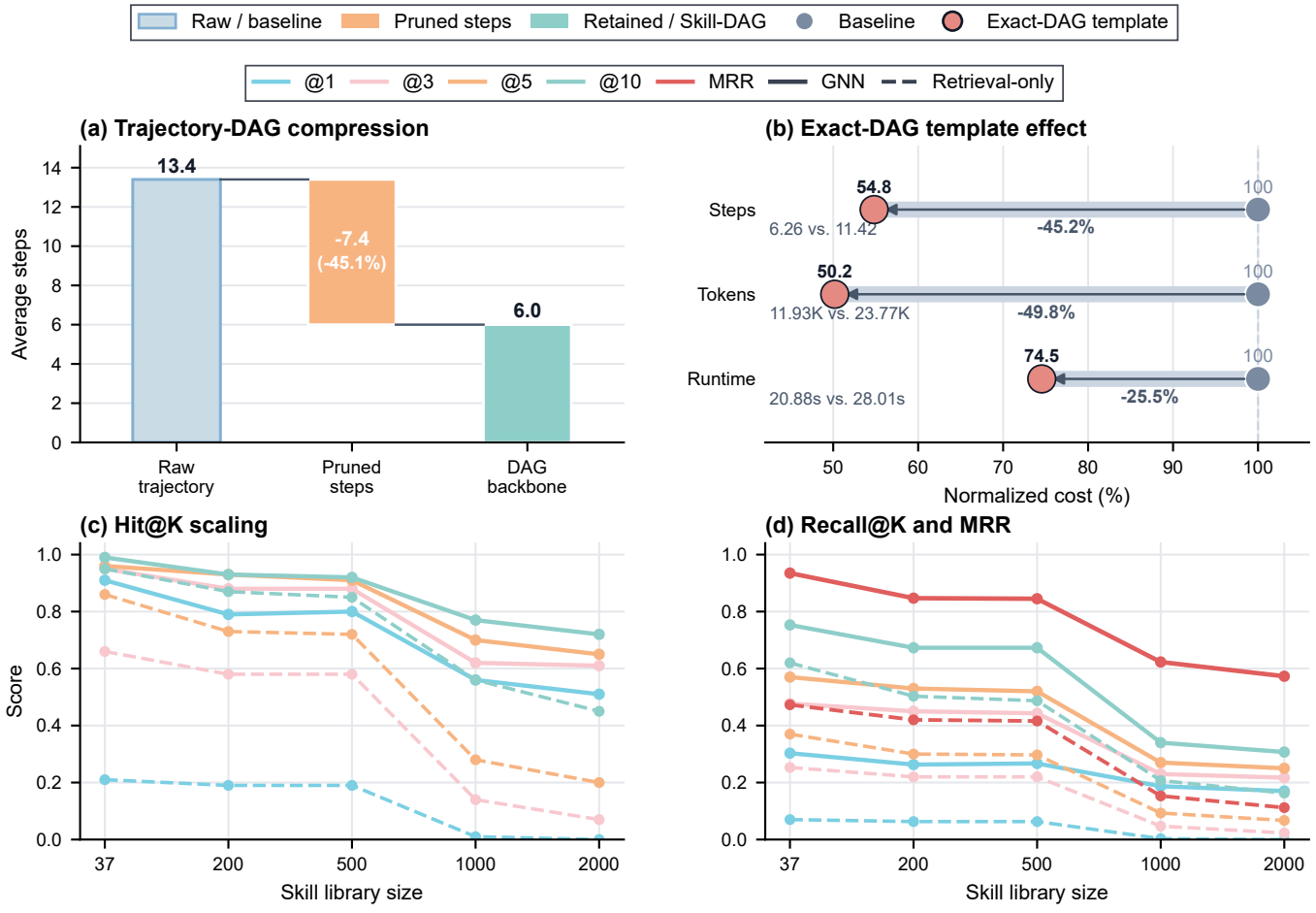

Raw / baseline
Pruned steps
Retained / Skill-DAG
Baseline
Exact-DAG template
@1
@3
@5
@10
MRR
GNN
Retrieval-only
(a) Trajectory-DAG compression
Average steps
13.4
-7.4
(-45.1%)
6.0
Raw trajectory
Pruned steps
DAG backbone
(b) Exact-DAG template effect
Steps
54.8
100
6.26 vs. 11.42
-45.2%
Tokens
50.2
100
11.93K vs. 23.77K
-49.8%
Runtime
74.5
100
20.88s vs. 28.01s
-25.5%
Normalized cost (%)
(c) Hit@K scaling
Score
Skill library size
(d) Recall@K and MRR
Skill library size